MADDRNet: A Morphology-Aware Dual-Domain Resonance Network for Remote Sensing Change Detection

Hanming Wei, Yujiang He, Lu Bai, Yuanxu Wang, Xin Yang, Junyi Lv, Yurong Qian, *Member, IEEE*, and Weijun Gong

***Abstract*—Remote sensing change detection (RSCD) under environmental interference, such as climatic variations and heterogeneous landforms, remains a challenging problem. Environmental variations often induce spectral style drift, which makes it difficult for existing methods to distinguish true semantic changes from pseudochanges. Meanwhile, anisotropic targets (e.g., winding rivers or fragmented farmland) pose additional challenges for morphology preservation, since rigid convolutional sampling tends to produce discontinuous structural representations. To address the representation dilemma between semantic generalization and fine-grained morphology preservation, we propose a morphology-aware dual-domain resonance network (MADDRNet), a novel semantic–morphology collaborative framework. Unlike conventional Siamese feature extraction architectures, the proposed framework integrates a frozen vision foundation model (VFM) to construct an environment-invariant macrolevel semantic prior, while a learnable branch focuses on capturing fine-grained structural details. To mitigate spectral style drift caused by environmental variations, a dual-domain resonance module (DDRM) is introduced to decouple semantic representations from climate-induced appearance discrepancies by integrating frequency-domain spectral gating with spatial-domain edge-aware feature injection. Furthermore, a morphology-aware gated fusion (MAGF) module dynamically adapts its receptive field to irregular target boundaries while suppressing nonsemantic background noise through temporal interaction. Extensive experiments on four benchmark datasets—CLCD, GFSWCLCD, LuojiaSET-CLCD, and Water-CD—demonstrate that the proposed method achieves state-of-the-art performance under diverse environmental conditions. This work provides a promising solution for robust RSCD under complex environmental interference and offers a new perspective for jointly modeling semantic generalization and fine-grained morphology preservation.**

***Index Terms*—Anisotropic targets, change detection, remote sensing, spectral style drift, vision foundation models (VFMs).**

Received 3 April 2026; revised 5 June 2026; accepted 20 June 2026. Date of publication 23 June 2026; date of current version 1 July 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62266043 and Grant 62566021, in part by the Excellent Youth Foundation of Xinjiang Uygur Autonomous Region of China under Grant 2023D01E01, in part by the Outstanding Young Talent Foundation of Xinjiang Uygur Autonomous Region of China under Grant 2023TSYCCX0043, in part by the Finance Science and Technology Project of Xinjiang Uyghur Autonomous Region under Grant 2023B01029-1 and Grant 2023B01029-2, and in part by the Natural Science Foundation of Gansu Province Science and Technology Plan under Grant 25JRRG037. *(Corresponding author: Yurong Qian.)*

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Digital Object Identifier 10.1109/TGRS.2026.3706593

I. Introduction

REMOTE sensing change detection (RSCD) is a fundamental technique for monitoring dynamic changes on the Earth's surface, with critical applications in precision agriculture [1], water resource management [2], disaster response [3], and urban expansion monitoring [4]. With the increasing spatial resolution and revisit frequency of Earth observation imagery, RSCD is expected not only to determine whether changes occur but also to delineate changed regions with clear boundaries and reliable semantic consistency [5], [6]. However, environmental interference, such as seasonal shifts, atmospheric variations, and illumination differences, often introduces significant spectral style drift between bi-temporal images, making it difficult to distinguish true land-cover changes from pseudochanges [7], [8]. This problem becomes more serious when the changed objects are anisotropic and complex (e.g., winding rivers, narrow shorelines, roads, and fragmented farmland parcels) [9], where semantic consistency and structural continuity must be preserved simultaneously.

In recent years, existing methods of RSCD can be broadly categorized into three groups: traditional approaches, deep-learning-based methods, and VFM-based methods. Traditional methods mainly rely on handcrafted features and mathematical transformations, such as image differencing [10], change vector analysis [11], and postclassification comparison [12]. Because they lack high-level semantic understanding and explicit cross-temporal contextual modeling, they have limited ability to maintain semantic consistency under seasonal or climatic variation. Moreover, their pixelwise comparison paradigm is often insufficient for distinguishing genuine land-cover transitions from environment-induced appearance variation, and it also makes accurate delineation of irregular change regions difficult [13]. Therefore, traditional methods do not adequately address environment-robust

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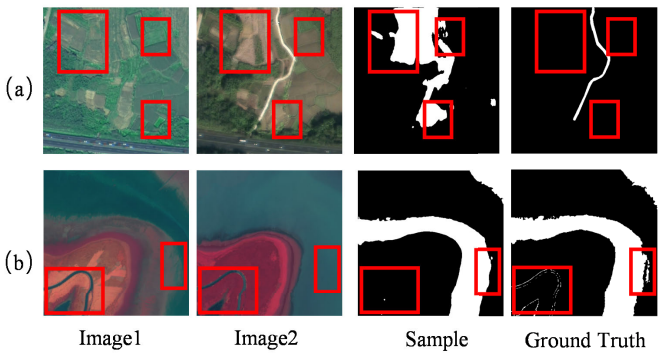


Fig. 1. Challenges: (a) Nonlinear spectral style drift induces pseudochanges interference and (b) there is a morphology mismatch between the receptive field and the true target boundary.

semantic discrimination, pseudochange suppression, cross-temporal consistency, or morphology preservation in complex remote sensing scenes. Deep learning has substantially advanced RSCD by enabling hierarchical representation learning and end-to-end optimization. Siamese CNNs, encoder–decoder architectures, and Transformer-based methods improve semantic abstraction, multiscale feature extraction, and bitemporal interaction, thereby alleviating part of the limitations of traditional approaches [14], [15]. Nevertheless, CNN-based methods are often constrained by local receptive fields and may not sufficiently capture long-range temporal dependencies [16], whereas globally modeling architectures may weaken local morphology details or remain sensitive to temporal appearance variation [17]. As a result, although deep learning methods improve change representation, their ability to simultaneously suppress pseudochanges [Fig. 1(a)] and preserve fine morphology remains limited [Fig. 1(b)]. VFM-based methods have opened a new direction for RSCD by transferring large-scale pretrained priors. By leveraging frozen foundation features [18], vision-language priors [19], or promptable segmentation knowledge [20], these methods provide stronger macrolevel semantics and improved generalization across scenes. However, most VFMs are pretrained on natural-image corpora rather than bitemporal remote sensing data, and their priors are not explicitly optimized for environmentally induced appearance variation, change-specific temporal discrimination, or morphology-sensitive localization [21]. Consequently, although VFM-based approaches strengthen semantic priors, they still fall short in jointly suppressing pseudochanges and preserving detailed target morphology in complex remote sensing scenes.

To address the above limitations, we propose a morphology-aware dual-domain resonance network (MADDRNet), a semantic–morphology collaborative framework for RSCD under environmental interference. Unlike conventional Siamese feature extraction architectures, the proposed network introduces the frozen VFM SkySense [22] to provide environment-invariant macrolevel semantic priors, while a learnable branch MobileNetV2 [23] is dedicated to capturing fine-grained morphology details. Furthermore, a dual-domain resonance module (DDRM) is designed to decouple stable semantic content from climate-induced appearance discrepancy by integrating frequency-domain spectral gating and spatial-domain edge-aware feature injection. A morphology-aware gated fusion (MAGF) module is then employed to adapt receptive fields to irregular target boundaries and suppress nonsemantic background noise through temporal interaction. The main contributions of this work are summarized as follows.

1) To mitigate pseudochanges caused by spectral style drift under environmental interference, we design a DDRM, which jointly exploits frequency-domain spectral gating and spatial-domain edge-aware feature injection to enhance environment-robust semantic representations.
2) To preserve morphological continuity for anisotropic and irregular targets, we propose a MAGF module, which dynamically adapts receptive fields to complex target boundaries while suppressing nonsemantic temporal noise.
3) To address the representation dilemma between semantic generalization and fine-grained morphology preservation in RSCD, we propose MADDRNet, a semantic–morphology collaborative framework that integrates a frozen VFM branch and a lightweight MobileNetV2 branch to jointly provide environment-invariant semantic priors and morphology-preserving structural details.

The rest of this article is organized as follows. Section II reviews related work on RSCD, including deep learning methods, VFMs, and frequency-domain strategies. Section III presents the proposed MADDRNet and its key components. Section IV describes the experimental setup and reports quantitative and ablation results. Section V provides further analysis, including interpretability and limitations. Finally, Section VI concludes this article and discusses future directions.

II. Related Works

A. Deep Learning Methods for Change Detection

Existing research in this field is mainly based on CNN- and Transformer-based architectures [24]. CNN-based methods are effective in capturing local spatial features. For example, BiUNet-Dense [25] improves fine-grained local representation by embedding Dense blocks into a U-Net backbone for bitemporal feature extraction. Building on this, later studies introduced Siamese architectures to explicitly model temporal correspondence and feature differences. EHCTNet [26] further employs a Siamese CNN-based extractor to learn multiscale local representations from paired temporal inputs, improving temporal interaction modeling in complex scenes. In addition, representative CNN-based methods such as GASNet [27] further enhance local change representation through multiscale feature aggregation. However, CNN-based methods are still limited by restricted receptive fields, which constrain long-range dependency modeling and global semantic perception. To overcome this limitation, Transformer-based methods were introduced into RSCD. By using self-attention, they improve

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long-range dependency modeling and cross-temporal semantic interaction [28]. CE-CDNet [29] enhances global change representation through Transformer-based channel optimization, while GRAD-Former [30] further combines global attention with gradient-guided refinement to better characterize structurally complex change regions. Nevertheless, token-based aggregation may still weaken local boundary sensitivity and fine morphology perception. More importantly, both CNN- and Transformer-based methods still rely on rigid geometric sampling, such as fixed convolution kernels or regular patch partitioning. This limits their adaptability to irregular and anisotropic target boundaries, often causing boundary distortion, incomplete delineation, and degraded morphology preservation in complex change regions.

B. Vision Foundation Models

VFMs have shown strong capability in learning high-level semantic representations from large-scale visual pretraining, offering new opportunities for RSCD. Early studies mainly focused on transferring such generalized priors to change detection under challenging conditions. RobSense [31] improves robustness to incomplete observations through spatio-temporal alignment and masked reconstruction. Extending this idea, panopticon [32] enables unified representation learning across heterogeneous remote sensing modalities. To better adapt pretrained semantics to change detection, RSPrompter [33] introduces automatic visual prompt generation, while Uni-Adapter [34] uses lightweight adapters for efficient task transfer. However, despite improved semantic generalization, VFM-based representations remain coarse-grained and are not well suited to fine-grained change localization. This issue is more evident for anisotropic and fragmented targets, where both global semantics and detailed morphology are required.

C. Frequency-Domain Learning for Visual Representation

Frequency-domain analysis provides an effective way to mitigate environmental noise and spectral style drift in RSCD. Since low-frequency components mainly encode global style statistics, early studies used frequency-domain representations to reduce illumination and climatic inconsistencies [35]. Building on this idea, FeaSpect [7] improves style-invariant representation by dynamically aligning amplitude spectra, while SFCNet [36] further enhances noise suppression through spatial-frequency interaction. To better separate invariant background information from change-related details, SAM-FDN [37], [38] explicitly decouples high-frequency changes from low-frequency stable components. However, in complex scenes with severe nonlinear spectral style drift, style noise and semantic content are often strongly coupled in the frequency space [39]. Under this condition, directly enforcing frequency-domain alignment may cause oversuppression of discriminative details, leading to blurred edges and incomplete morphology preservation [40], [41]. Therefore, achieving effective frequency-domain noise suppression without sacrificing fine-grained morphology remains a critical challenge.

Algorithm 1 Forward Pass of MADDRNet

Input: Bi-temporal images T_1, T_2
Output: Final change map $M = M_1$
for $j \in \{1, 2\}$ **do**
  Extract multi-scale features from T_j using MobileNetV2:
    $F_1^{\text{mob},T_j}, \ldots, F_5^{\text{mob},T_j} = \text{MobileNetV2}(T_j)$
  Extract macro-level semantic features from SkySense (frozen VFM):
    $F_1^{\text{VFM},T_j}, \ldots, F_4^{\text{VFM},T_j} = \text{SkySense}(T_j)$
  Fuse VFM and MobileNet features via bridging modules:
    $F_1^{T_j} = F_1^{\text{mob},T_j}$
    $F_2^{T_j} = \text{Bridge}\big(F_2^{\text{mob},T_j}, F_1^{\text{VFM},T_j}\big)$
    $F_3^{T_j} = \text{Bridge}\big(F_3^{\text{mob},T_j}, F_2^{\text{VFM},T_j}\big)$
    $F_4^{T_j} = \text{Bridge}\big(F_4^{\text{mob},T_j}, F_3^{\text{VFM},T_j}\big)$
    $F_5^{T_j} = \text{Bridge}\big(F_5^{\text{mob},T_j}, F_4^{\text{VFM},T_j}\big)$
  Enhance features with DDRM modules:
    $E_1^{T_j}, \ldots, E_5^{T_j} = \text{DDRM}(F_1^{T_j}, \ldots, F_5^{T_j})$
end for
Fuse bi-temporal features using MAGF units:
  $C_i = \text{MAGF}(E_i^{T_1}, E_i^{T_2}),\ i = 1, \ldots, 5$
Decode multi-scale features into hierarchical decoder representations:
  $L_4 = \text{C-B-R}\big(\text{Concat}\big(\text{Upsample}(C_5), C_4\big)\big)$
  $L_3 = \text{C-B-R}\big(\text{Concat}\big(\text{Upsample}(L_4), C_3\big)\big)$
  $L_2 = \text{C-B-R}\big(\text{Concat}\big(\text{Upsample}(L_3), C_2\big)\big)$
  $L_1 = \text{C-B-R}\big(\text{Concat}\big(\text{Upsample}(L_2), C_1\big)\big)$
Transform decoder representations into prediction maps:
  $M_s = \text{Sigmoid}\big(\text{Upsample}(L_s)\big),\ s = 1, 2, 3, 4$
return M_1

III. Methodology

In this section, we first present the overall architecture of the proposed network. Then, we describe the semantic–morphology collaborative framework and the interaction between the frozen VFM and the lightweight branch. The DDRM and MAGF modules are introduced for spectral style drift mitigation and morphology-aware adaptation, respectively. Finally, we describe the loss function used to optimize MADDRNet.

A. Overall Structure of MADDRNet

The overall architecture of MADDRNet is illustrated in Fig. 2. The MADDRNet forward pass first extracts multiscale local features using MobileNetV2 and environment-invariant semantic features with the frozen SkySense VFM. Bridging modules fuse the two feature sets at each scale. DDRM suppresses spectral style drift, and MAGF further refines the features by dynamically adapting receptive fields to irregular and anisotropic target boundaries, effectively preserving fine-grained morphology. Finally, the multiscale features are decoded by C-B-R blocks into four hierarchical decoder representations, denoted as L_1, L_2, L_3, and L_4 (where C-B-R denotes Conv, BatchNorm, and ReLU). These representations are further upsampled and passed through sigmoid activations

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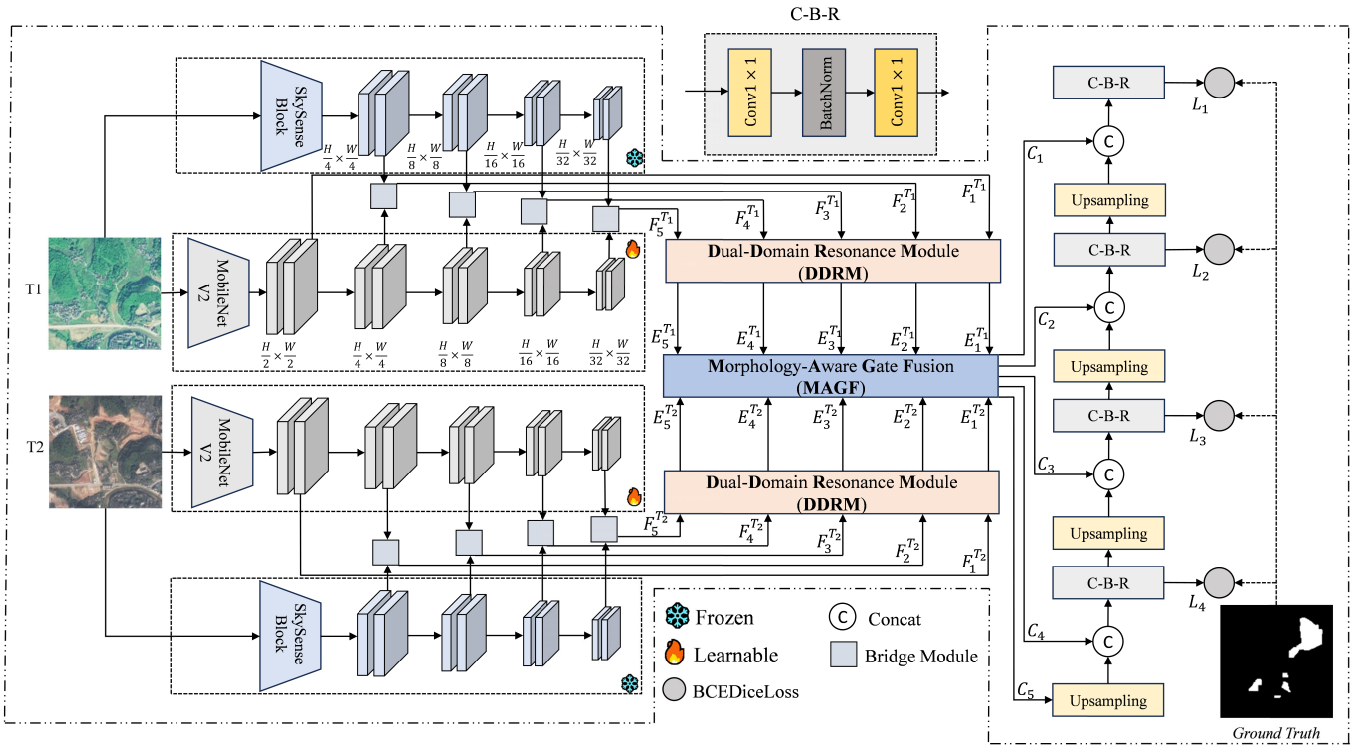


Fig. 2. Overview of the MADDRNet framework. It integrates a pretrained SkySense and a learnable MobileNetV2 as the backbone for feature extraction, followed by the DDRM for anti-interference and MAGF for boundary refinement and feature fusion. The decoder generates four hierarchical representations L_1–L_4, which are transformed into multiscale prediction maps by upsampling and sigmoid activation for BCE-Dice supervision. The highest resolution prediction is used as the final change map during inference. For compactness, the upsampling and sigmoid operations before the supervised prediction maps are omitted in the schematic. The bridge module is adapted from BAN [18].

to generate multiscale prediction maps for auxiliary supervision. During inference, only the highest resolution prediction map is used as the final change map. The pseudocode for the overall structure is shown in Algorithm 1.

B. Semantic–Morphology Collaboration Framework

Traditional CNN-based methods face a tradeoff between semantic abstraction and detail preservation, while VFMs provide strong semantics but are expensive to fine-tune. Therefore, we design a low-cost semantic–morphology collaborative architecture.

As illustrated in Fig. 2, the framework consists of two complementary branches. The frozen SkySense branch serves as the semantic branch, providing robust environment-invariant semantic priors from large-scale pretraining. It remains fixed during training to preserve its generalization capability while avoiding the high cost of full fine-tuning. The resulting multiscale semantic features are denoted as F_i^{VFM}, where $i \in \{1, 2, 3, 4\}$. In parallel, a lightweight MobileNetV2 is employed as the morphology branch to capture fine-grained local structures and texture details that are easily smoothed out in frozen VFM representations. Its output features are denoted as F_i^{mob}. To enable semantic–morphology collaboration, we adopt the bridging module from BAN [18] to align F_i^{VFM} with the feature space of F_i^{mob}, allowing robust semantic priors to guide the learning of fine-grained morphology representations.

We perform layer normalization (LN) and linear dimensionality reduction on F_i^{VFM} to eliminate numerical instabilities and align channel dimensions. Then, F_i^{mob} is used as the query vector to construct the cross-modal affinity matrix. Finally, through the double residual, the original details, the global semantic prior and the filtered semantic features are fused to generate the final hybrid feature F_i. The above process is as follows:

$$\tilde{F}_i^{\mathrm{VFM}} = \phi\left(\mathrm{LN}\left(F_i^{\mathrm{VFM}}\right)\right) \tag{1}$$

$$F_i^{att} = \mathrm{Softmax}\left(\frac{F_i^{\mathrm{mob}}\left(\tilde{F}_i^{\mathrm{VFM}}\right)^{\top}}{\sqrt{d}}\right) \cdot \tilde{F}_i^{\mathrm{VFM}} \tag{2}$$

$$F_i = F_i^{\mathrm{mob}} + \mathcal{T}\left(\tilde{F}_i^{\mathrm{VFM}}\right) + F_i^{att} \tag{3}$$

where $\mathrm{LN}(\cdot)$ represents layer normalization, $\phi(\cdot)$ denotes the linear projection layer, d is the scaling factor, and $\mathcal{T}(\cdot)$ denotes bilinear interpolation. This design leverages LN, linear projection, and bilinear interpolation to preserve sensitivity to fine-grained changes while maintaining the semantic generalization provided by the VFM.

C. Dual-Domain Resonance Module

Fig. 3 illustrates the architecture of DDRM. It is designed to alleviate the confusion between true semantic changes and environment-induced pseudochanges. By suppressing spectral style drift and compensating for detail loss, DDRM enhances the robustness of feature representations under seasonal and illumination variations.

Given the bridged feature $F_i \in \mathbb{R}^{C \times H \times W}$ at the ith scale, DDRM first maps it into a unified representation through a

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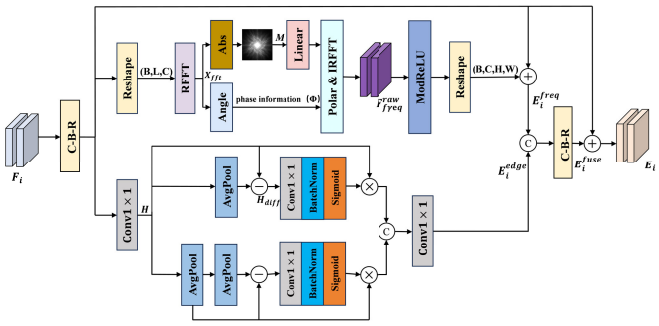


Fig. 3. Architecture of the DDRM. The module works through the synergy of spectral gating in the frequency domain with edge injection in the spatial domain, which mitigates spurious variation interference caused by nonlinear spectral style drift.

1×1 convolution, and then processes it in parallel in the frequency and spatial domains. The overall formulation can be written as

$$F_i' = \text{C-B-R}\left(F_i\right)$$
$$E_i = \text{DDRM}\left(F_i\right) = \mathcal{G}\left(\mathcal{F}_{freq}\left(F_i'\right), \mathcal{F}_{edge}\left(F_i'\right)\right) \tag{4}$$

where F_i' denotes the transformed feature, E_i denotes the output of DDRM, $\mathcal{F}_{freq}(\cdot)$ and $\mathcal{F}_{edge}(\cdot)$ denote the two parallel transformations, and $\mathcal{G}(\cdot)$ denotes the fusion operation.

To mitigate spectral style drift, F_i' is first decomposed into amplitude and phase spectra by the 2-D real fast Fourier transform (RFFT)

$$X_{fft} = \mathcal{F}\left(F_i'\right). \tag{5}$$

This complex-valued spectrum is decomposed into its amplitude and phase components, corresponding to the "Abs" and "Angle" blocks in Fig. 3

$$M = \left|X_{fft}\right|, \quad \Phi = \angle X_{fft} \tag{6}$$

where $M \in \mathbb{R}^{C\times H\times W}$ denotes the amplitude spectrum and $\Phi \in \mathbb{R}^{C\times H\times W}$ denotes the phase spectrum. Here, $|\cdot|$ represents the modulus of a complex number, while $\angle(\cdot)$ denotes its phase angle, computed from the real and imaginary parts.

The amplitude spectrum M is first processed by a learnable linear layer parameterized by $W_{\text{freq}} \in \mathbb{R}^{C\times C}$

$$\widetilde{M} = W_{\text{freq}} M. \tag{7}$$

The filtered amplitude $\widetilde{M}$ is then recombined with the original phase Φ in the *Polar and IRFFT* block and transformed back to the spatial domain

$$F_{freq}^{raw} = \mathcal{F}^{-1}\left(\widetilde{M} \cdot e^{i\Phi}\right). \tag{8}$$

Finally, a ModReLU [42] activation is applied with a weighted residual connection

$$E_i^{freq} = F_i' + \alpha \cdot \text{ModReLU}\left(F_{freq}^{raw}\right) \tag{9}$$

where α is a learnable residual weight initialized to 0.5.

Since frequency filtering may weaken local boundaries, spatial detail compensation is further introduced. For a dimension-reduced feature H, the local high-frequency response is extracted by

$$H_{diff} = H - \text{AvgPool}_{3\times 3}\left(H\right). \tag{10}$$

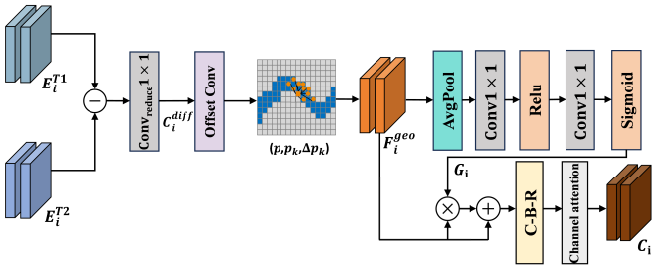


Fig. 4. MAGF architecture integrates anisotropic receptive field adaptation and temporal interactive gating, and accurately fits irregular topological boundaries while suppressing nonsemantic environmental noise.

Based on this, an edge-aware attention map is generated to enhance structural details

$$E\left(H\right) = \text{Sigmoid}\left(\text{BN}\left(\text{Conv}_{1\times 1}\left(H_{diff}\right)\right)\right) \odot H \tag{11}$$

where $\text{AvgPool}_{3\times 3}(\cdot)$ denotes average pooling, BN denotes batch normalization, and $\odot$ denotes elementwise multiplication. To capture details at different smoothing levels, two edge responses are extracted and fused as

$$E_i^{edge} = \text{Conv}_{1\times 1}\left(\text{Concat}\left[E\left(H\right), \quad E\left(\text{AvgPool}\left(H\right)\right)\right]\right). \tag{12}$$

Finally, the two refined features are concatenated and fused, and the result is added back through a residual connection

$$E_i^{fuse} = \text{C-B-R}\left(\text{Concat}\left[E_i^{freq}, E_i^{edge}\right]\right) \tag{13}$$
$$E_i = F_i' + E_i^{fuse} \tag{14}$$

where E_i is the final output of DDRM at the ith scale. In this way, DDRM suppresses spectral style drift while preserving fine-grained morphology details.

D. MAGF Module

As shown in Fig. 4, we propose MAGF to address the limitations of spatial rigidity and feature rigidity faced by standard convolutions when dealing with anisotropic objects. Network flexibility is achieved through adaptive kernel convolution. The geometric-aware gating mechanism is introduced to give the network feature selection to achieve high-fidelity reconstruction of complex topology.

For the bitemporal input features $E_i \in \mathbb{R}^{B\times C\times H\times W}$, the module captures the initial change intensity by calculating the absolute difference between them. To reduce computation and focus information, a 1×1 convolution is used to compress the channel dimension and generate a compact difference feature map C_i^{diff}

$$C_i^{diff} = \text{Conv}_{reduce}\left(\left|E_i^{T1} - E_i^{T2}\right|\right) \tag{15}$$

where $|\cdot|$ denotes the elementwise absolute difference and Conv_{reduce} maps the number of channels from C to C_{mid}. To break spatial rigidity, unlike standard convolutions, adaptive sampling predicts coordinate offsets based on local feature gradients and dynamically adjusts receptive field shapes. For

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any pixel p on the feature map, the spatially aligned geometric feature F_i^{geo} is computed as follows:

$$F_i^{geo} = \sum_{k=1}^{K} w_k \cdot C_i^{diff}\left(p + p_k + \Delta p_k\right). \tag{16}$$

K represents the number of sampling points, w_k represents the learnable weights, and p_k is the initial regular grid coordinates. Δp_k is the learnable irregular offset.

In order to break the limited feature adaptability, the geometry-aware gating mechanism dynamically screens the best matching feature channels according to the current geometric morphology

$$G_i = \text{Sigmoid}\left(\text{Conv}_{ez}\left(\text{ReLU}\left(\text{Conv}_{sq}\left(\text{AvgPool}\left(C_i^{geo}\right)\right)\right)\right)\right) \tag{17}$$

$$C_i = \text{CAM}\left(\text{C-B-R}\left(F_i^{geo} + \left(F_i^{geo} \cdot \left(1 + G_i\right)\right)\right)\right) \tag{18}$$

where Conv_{sq} and Conv_{ez} constitute the bottleneck structure to model the interchannel dependencies. The term $(1 + G_i)$ implements geometric calibration, which dynamically amplifies the most relevant geometric feature channels based on the captured shape (e.g., enhancing curvilinear texture responses when dealing with rivers) while suppressing mismatched noise. CAM stands for channel attention module for final feature refinement.

E. Loss Function

MADDRNet is trained as a binary change segmentation model. As described in Algorithm 1, the decoder produces four hierarchical intermediate representations, denoted as L_1, L_2, L_3, and L_4. Each representation L_s is upsampled to the ground-truth resolution and passed through a sigmoid activation to obtain a prediction map M_s, where $s = 1, \ldots, 4$. The highest resolution prediction map M_1 is used as the final prediction during inference.

Let $M_s = \{m_{s,n}\}_{n=1}^{N}$ and $Y = \{y_n\}_{n=1}^{N}$ denote the sth sigmoid-normalized prediction map and the binary ground truth, respectively, where n is the pixel index and N is the total number of pixels. The BCE-Dice loss is used to supervise each prediction map

$$\mathcal{J}_s = \mathcal{L}_{\text{BCE}}\left(M_s, Y\right) + \mathcal{L}_{\text{Dice}}\left(M_s, Y\right). \tag{19}$$

The BCE loss is defined as

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{n=1}^{N}\left[y_n \log m_{s,n} + (1 - y_n)\log\left(1 - m_{s,n}\right)\right]. \tag{20}$$

The Dice loss is defined as

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_{n=1}^{N} m_{s,n} y_n + \epsilon}{\sum_{n=1}^{N} m_{s,n} + \sum_{n=1}^{N} y_n + \epsilon} \tag{21}$$

where ϵ is used to avoid numerical instability. The total loss is computed by summing the losses of four prediction maps

$$\mathcal{J}_{\text{total}} = \sum_{s=1}^{4} \mathcal{J}_s. \tag{22}$$

The BCE term provides pixelwise changed/unchanged supervision, while the Dice term alleviates the class imbalance caused by sparse changed regions. The multiscale supervision encourages the decoder to learn change-aware representations at different resolutions, improving both semantic consistency and boundary preservation.

IV. Experiments

This section evaluates the effectiveness and robustness of MADDRNet. Sections IV-A and IV-B introduce the benchmark datasets, evaluation metrics, and experimental setup. Section IV-C presents qualitative and quantitative comparisons with state-of-the-art methods. Section IV-D analyzes the training loss and convergence behavior. Section IV-E reports ablation studies analyzing the contributions of DDRM, MAGF, and the semantic–morphology collaboration paradigm.

A. Dataset Details

We evaluate MADDRNet on four datasets, including CLCD, GFSWCLCD, LuojiaSET-CLCD, and Water-CD. These datasets cover a variety of challenging scenes, from fine texture changes of cropland and urban building expansion to complex water bodies and multigeomorphic heterogeneous environments.

1) CLCD [43]: The dataset was acquired by the Gaofen-2 satellite and contains 600 pairs of dual-temporal images of size 512×512 pixels, whose spatial resolution covers 0.5 to 2 m. It mainly covers cropland areas in Guangdong, China, with a focus on the conversion of arable land to nonarable land. In order to ensure the fairness of the experiment, the dataset is divided into training, validation, and test sets according to 6:2:2.

2) GFSWCLCD [44]: The dataset was acquired by the Gaofen-2 satellite and is specifically constructed for detecting complex changes in cultivated land. It contains 1144 pairs of dual-temporal images of size 256×256 pixels with a spatial resolution of 1 m. The images cover diverse geographical environments, including plains and lakes in Tacheng, Xinjiang, China, and focus on capturing seasonal and structural changes of agricultural land, such as cropland expansion and abandonment. The dataset is divided into training, validation, and test sets in a 6:2:2 ratio.

3) LuojiaSET-CLCD [45]: The dataset is a large scale, complex change detection benchmark capturing heterogeneous scenes. It contains 4194 pairs of dual-temporal images of size 512×512 pixels with an optical resolution of 1–2 m. The images cover a variety of landforms, including urban areas, plains, mountainous regions, and arable land, and include numerous small-scale building changes and complex background interference. The dataset is randomly divided into training, validation, and test sets in a 7:1:2 ratio.

4) Water-CD [2]: The dataset was constructed from Sentinel-2 satellite imagery and focuses on detecting dynamic changes in water bodies. It contains 1149 pairs of images with a spatial resolution of 10 m. The images cover areas such as seasonal lakes in the Yangtze River basin, China, and the Jumna River, South Asia, spanning from 2016 to 2022.

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The dataset emphasizes water ebb and flow between the rainy and dry seasons, effectively evaluating the model's ability to capture periodic natural changes. The dataset is randomly divided into training, validation, and test sets in a 7:2:1 ratio.

B. Implementation Details and Metrics

1) Implementation Details: MADDRNet is implemented in PyTorch version 1.12.0, CUDA 11.6; all training and inference are performed on a single NVIDIA A40 GPU (48 GB) with a batch size of 8 and a total of 100 epochs. The Adam optimizer is used to update the network parameters with momentum parameters $\beta_1 = 0.9$, $\beta_2 = 0.99$, and weight decay of 1×10^{-4}. The initial learning rate is set to 5×10^{-4}. The Poly learning rate decay strategy is adopted to avoid falling into a local optimum; the input image is normalized, and the data is augmented by scaling, cropping, flipping, and channel swapping. The network is optimized using the multiscale BCE-Dice loss described in Section III-E. SkySense uses the original pretrained weights and remains frozen during training; MobileNetV2 and the rest of the core components of MADDRNet (including DDRM, MAGF Module, and decoder) are trained from scratch with a random initialization strategy, and all participating comparison methods are reimplemented under the same settings. Our source code will be publicly available upon acceptance of the article.

2) Evaluation Metrics: To quantitatively evaluate the change detection performance of MADDRNet, a set of standard evaluation metrics is used. These include precision (P), recall (R), $F1$-score ($F1$), intersection over union (IoU), and overall accuracy (OA). The calculation formulas of the main metrics are as follows:

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{23}$$

$$\mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{24}$$

$$\mathrm{F1} = \frac{2 \times \mathrm{P} \times \mathrm{R}}{\mathrm{P} + \mathrm{R}} \tag{25}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{26}$$

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{27}$$

where TP, FP, TN, and FN denote true positives, false positives, true negatives, and false negatives, respectively. Higher values indicate better performance.

C. Comparison Methods

To verify the superiority of MADDRNet, we comprehensively compare it with ten representative SOTA methods. These include pure CNN-based methods (B2CNet, CACDNet, and GASNet), Transformer-based methods (ELGCNet, FDFENet, and Farm-CD), and foundation model-based methods (SAM-CD, BAN, STMTNet, and NeXt2Former-CD). The details of each comparison method are as follows.

1) *B2CNet [46]:* A "bounding-to-center" progressive refinement strategy that utilizes a changing bounding-aware module to address boundary uncertainty in feature difference.
2) *CACDNet [47]:* By adopting a dual-decoder architecture, the high-level semantics are utilized to locate the changed areas, and the low-level features are utilized to refine the texture details, thereby achieving precise generation of the changed images.
3) *GASNet [27]:* The global awareness Siamese network structure is introduced, and the global attention mechanism and foreground awareness strategy are used to capture long-distance dependencies to alleviate the problems of class imbalance and scale change.
4) *ELGCNet [48]:* A local–global context aggregation mechanism is proposed, and pooled transposed attention is used to achieve robust feature extraction while being lightweight.
5) *FDFENet [49]:* By combining frequency domain analysis and Transformer, the content and style information of the image are decoupled through the frequency domain feature exchange module, and the robustness of the model to seasonal differences is enhanced.
6) *Farm-CD [50]:* An agricultural information-guided gated network that focuses on farmland-related changes while suppressing nonfarmland interference through farmland-aware feature extraction and gated refinement.
7) *SAM-CD [51]:* The application of SAM in change detection, through parameter fine-tuning and feature interaction, transfers SAM's powerful generalization segmentation ability to the change detection task.
8) *BAN [18]:* The frozen pretrained base model is used to extract generic features and inject domain knowledge through adapters to achieve efficient cross-domain transfer.
9) *STMTNet [52]:* A spatio-temporal multiscale ternary network is designed for farmland change detection, which combines FastSAM and VGGNet16, and effectively suppresses environmental noise such as cloud occlusion through spatio-temporal adaptive gating.
10) *NeXt2Former-CD [53]:* A vision foundation model (VFM)-based change detection framework that employs a Siamese DINOv3-pretrained ConvNeXt encoder, deformable attention-based temporal fusion, and a Mask2Former decoder for robust bi-temporal feature modeling.

1) Quantitative Evaluation: Tables I and II summarize the performance of all compared methods on the key metrics P, R, $F1$, IoU, and OA. The details are as follows.

a) Performance on the CLCD dataset: As shown in Table I, MADDRNet achieves an $F1$ score of 83.39% and an IoU of 71.51%, demonstrating clear advantages in cropland change detection under strong seasonal and phenological variations. In particular, compared with FDFENet, which also introduces frequency-domain modeling to mitigate phenological discrepancies, MADDRNet improves $F1$ and IoU by 6.30% and 8.79%, respectively. MADDRNet also surpasses STMTNet, a strong method tailored for farmland scenarios, by 9.47% in $F1$. These results indicate that the proposed architecture can more effectively suppress pseudochanges caused by appearance shifts while preserving discriminative change cues in agricultural scenes.

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TABLE I
QUANTITATIVE COMPARISON RESULTS ON CLCD, GFSWCLCD, AND LUOJIASET-CLCD DATASETS

Method	CLCD					GFSWCLCD					LuojiaSET-CLCD				
	P (%)	R (%)	F1 (%)	IoU (%)	OA (%)	P (%)	R (%)	F1 (%)	IoU (%)	OA (%)	P (%)	R (%)	F1 (%)	IoU (%)	OA (%)
GASNet	63.17	75.51	68.79	52.42	94.90	81.35	83.80	82.55	70.29	90.62	67.87	77.49	73.36	56.70	98.01
B2CNet	65.48	78.80	71.53	55.67	95.33	80.30	83.50	81.87	69.30	90.20	60.66	86.23	71.22	55.30	97.66
CACDNet	71.91	81.09	76.23	61.58	96.24	84.26	80.44	82.31	69.94	90.74	70.13	78.03	73.87	58.57	98.15
ELGCNet	71.20	61.12	65.78	49.07	95.27	83.26	76.41	79.68	66.22	89.68	72.65	68.67	70.60	54.56	98.08
FDFENet	81.08	73.47	77.09	62.72	96.75	83.07	86.06	84.54	73.22	91.66	74.02	74.43	74.23	59.02	98.26
Farm-CD	74.81	73.69	74.24	59.04	96.19	81.70	76.11	78.80	65.02	89.15	67.80	59.43	63.36	46.37	97.67
SAM-CD	68.96	74.16	71.47	55.60	95.57	82.78	79.91	81.32	68.52	90.38	78.35	72.50	75.31	60.40	98.40
BAN	68.40	69.80	69.05	51.28	94.20	82.38	78.32	80.30	67.09	89.82	78.82	72.65	75.61	60.78	97.65
STMTNet	69.79	78.57	73.92	58.63	95.85	82.24	83.42	83.83	72.16	91.57	65.11	79.78	71.70	55.88	97.88
NeXt2Former-CD	79.68	82.26	80.95	68.03	97.12	85.15	81.99	83.54	71.95	91.53	79.89	75.75	77.76	63.62	98.54
MADDRNet (Ours)	85.61	81.28	83.39	71.51	97.59	85.21	89.35	87.23	77.36	93.07	76.87	80.52	78.65	64.82	98.53

Note: Red indicates the best performance, Blue indicates the second-best performance.

TABLE II
QUANTITATIVE COMPARISON RESULTS ON WATER-CD DATASET

Method	Backbone	Params (M)	FLOPs (G)	Water-CD				
				P (%)	R (%)	F1 (%)	IoU (%)	OA (%)
GASNet	CNN	28.91	306.93	90.84	91.49	91.16	83.76	96.75
B2CNet	CNN	16.10	22.36	85.82	93.50	89.49	80.99	95.98
CACDNet	CNN	28.75	53.72	88.23	86.83	87.52	77.81	95.44
ELGCNet	Transformer	10.57	120.69	88.73	88.31	88.52	79.40	95.80
FDFENet	Transformer	90.55	33.93	91.47	88.96	90.20	82.15	96.46
Farm-CD	Transformer	48.79	32.13	87.37	83.03	85.14	74.13	94.69
SAM-CD	VFM	8.60	2.59	91.20	88.70	89.94	81.71	96.36
BAN	VFM	68.88	29.21	91.50	89.71	90.59	82.80	89.72
STMTNet	VFM	60.25	31.79	92.27	88.95	90.58	82.78	96.61
NeXt2Former-CD	VFM	392.00	159.54	87.46	90.83	89.11	80.35	95.93
MADDRNet (Ours)	VFM	79.81	41.13	92.55	90.36	91.45	84.24	96.90

Note: Red indicates the best performance, Blue indicates the second-best performance.

b) Performance on the GFSWCLCD dataset: MADDRNet consistently outperforms representative lightweight CNN and Transformer-based methods, confirming its effectiveness in balancing efficiency and accuracy. In particular, compared with the Transformer-based ELGCNet, MADDRNet achieves gains of 7.55% in $F1$ and 11.14% in IoU. In addition, it improves OA by 2.33% over CACDNet, a strong CNN-based baseline. These results suggest that MADDRNet is better able to maintain global semantic consistency while retaining local structural details, which is particularly important in complex farmland scenes with subtle and fragmented changes.

c) Performance on the LuojiaSET-CLCD dataset: MADDRNet achieves the best $F1$-score and IoU on the LuojiaSET-CLCD dataset while maintaining a comparable OA of 98.53%, as reported in Table I. Notably, compared with recent VFM-based methods such as SAM-CD and BAN, MADDRNet improves $F1$ by 3.34% and 3.04%, respectively, indicating that the proposed framework can exploit foundation-level semantics more effectively for change detection. Moreover, against FDFENet and GASNet, MADDRNet yields gains of 4.42%/5.80% and 6.29%/8.12% in $F1$/IoU, respectively. Although MADDRNet does not obtain the highest precision or recall, LuojiaSET-CLCD contains sparse small targets and cluttered backgrounds, where recall-oriented predictions may introduce more FP, while precision-oriented predictions may miss ambiguous small changes. MADDRNet achieves a better balance between false alarms and missed detections, leading to the best $F1$ and IoU with comparable OA. These consistent improvements verify the robustness of MADDRNet in complex scenes, where confusing background patterns often lead to both false alarms and missed detections.

d) Performance on the Water-CD dataset: As shown in Table II, MADDRNet achieves an $F1$ score of 91.45% and an IoU of 84.24%, showing strong capability in delineating water-body changes with clear and complete boundaries. Compared with B2CNet and CACDNet, two methods that emphasize boundary refinement, MADDRNet improves $F1$ by 1.95% and 3.25%, and IoU by 3.93% and 6.43%, respectively. For the Water-CD dataset, MADDRNet does not achieve the highest recall because it avoids overexpanding predictions in blurred water–land transition areas. Although B2CNet obtains higher recall, its lower precision and $F1$ indicate more FP. On the contrary, MADDRNet better balances boundary accuracy and false-alarm suppression, leading to more reliable overall performance.

2) Qualitative Evaluation: To visually evaluate the model performance, Figs. 5–8 show the visual comparison results of MADDRNet and other SOTA methods on typical samples. Different colors are used to distinguish the correctness or incorrectness of the results, including TP (white), FP (green), TN (black), and FN (red).

a) Advantages in reducing pseudochange: Spectral style drift caused by the environment is a difficult problem in RSCD. As shown in s1, s2, and s3 of Fig. 5 and s3 of Fig. 6, the vegetation in the bitemporal images presents severe spectral variations, and the traditional GASNet and B2CNet tend to misjudge these unstructured phenological differences as real changes, resulting in large green false-alarm regions

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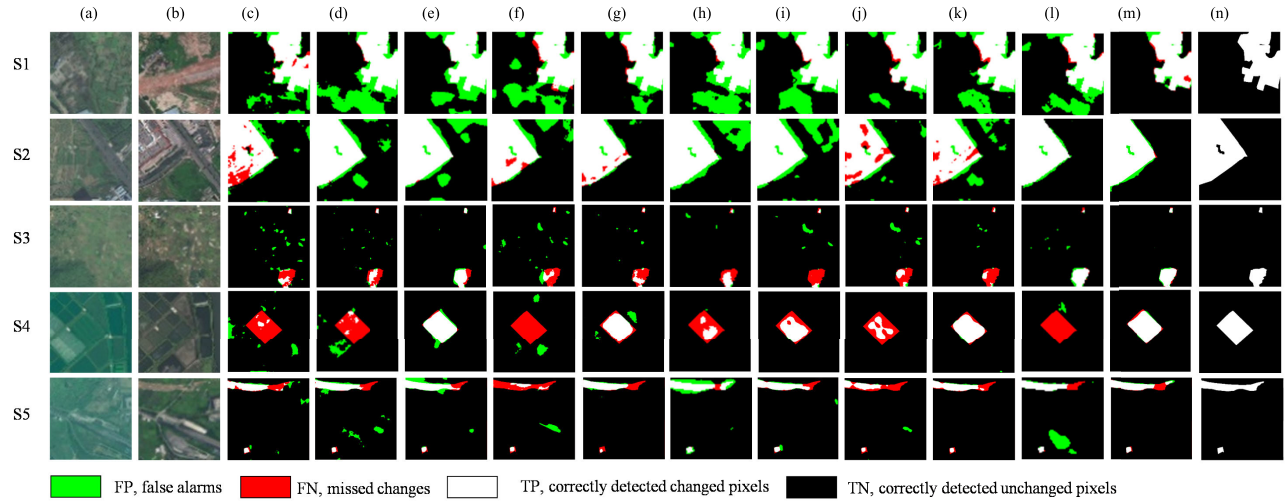


Fig. 5. Visual comparison results of the proposed method and SOTA methods on the CLCD dataset. S1–S5 Five representative bitemporal samples selected from typical cropland scenes with seasonal appearance variations, fragmented parcels, and small scattered change regions. (a) Image1. (b) Image2. (c) GASNet. (d) CACDNet. (e) B2CNet. (f) ELGCNet. (g) FDFENet. (h) Farm-CD. (i) SAM-CD. (j) BAN. (k) STMTNet. (l) NeXt2Former-CD. (m) MADDRNet. (n) GT. The color legend of the detection maps is as follows: green denotes FP, red denotes FN, white denotes TP, and black denotes TN.

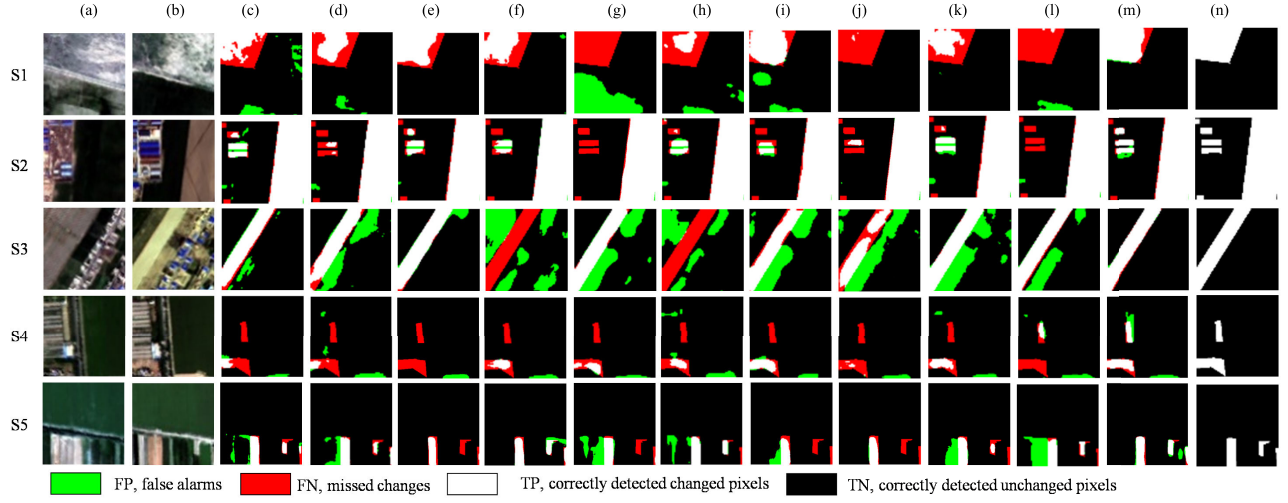


Fig. 6. Visual comparison results of the proposed method and SOTA methods on the GFSWCLCD dataset. S1–S5 Five representative samples containing complex farmland backgrounds, irregular change regions, and ambiguous local details. (a) Image1. (b) Image2. (c) GASNet. (d) CACDNet. (e) B2CNet. (f) ELGCNet. (g) FDFENet. (h) Farm-CD. (i) SAM-CD. (j) BAN. (k) STMTNet. (l) NeXt2Former-CD. (m) MADDRNet. (n) GT. The color legend of the detection maps is as follows: green denotes FP, red denotes FN, white denotes TP, and black denotes TN.

in the predicted maps. SAM-CD is overly sensitive to low-level texture semantics; consequently, it struggles to filter out seasonal noise and simultaneously produces a substantial number of FPs. MADDRNet uses the dual-domain cooperative operation of DDRM to successfully suppress the amplitude spectrum interference representing style changes and focus on the phase spectrum features representing the ground object structure. True changes are retained, and FP are reduced.

b) Advantages in boundaries and details: The shortcomings of existing methods are particularly obvious when dealing with small targets and irregular boundaries in complex scenes. As shown in s2, s4, s5 of Fig. 7 and s1, s2 of Fig. 8, the prediction results of ELGCNet and BAN show blurred or sticky boundaries. However, CACDNet introduces the edge perception mechanism; it still produces red missed-detection regions around small targets and cannot capture subtle change features. Similarly, water edges tend to be fragmented and meandering in the Water-CD dataset. s1, s2, s3, and s5 of Fig. 8 show that MADDRNet exhibits a "fluid-like" perception ability using MAGF. The boundary of the prediction map is clear and sharp, which closely fits the real topology of the ground objects. Compared with other methods, MADDRNet not only significantly reduces red missed-detection regions of small targets but also maintains high edge integrity at complex land–water junctions, and its visualization results are highly consistent with the ground truth.

3) Model Efficiency: As shown in Table II, MADDRNet achieves a favorable tradeoff between model complexity and detection performance. Compared with NeXt2Former-CD, which contains 392.00M parameters and requires 159.54G

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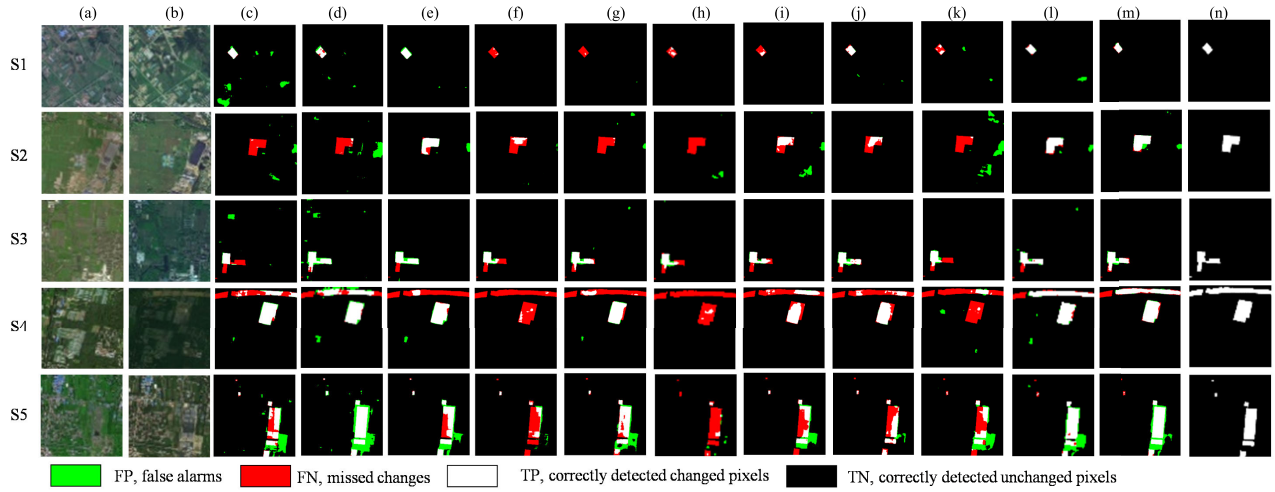


Fig. 7. Visual comparison results of the proposed method and SOTA methods on the LuojiaSET-CLCD dataset. S1–S5 Five representative samples with small targets, sparse change regions, and cluttered scene backgrounds. (a) Image1. (b) Image2. (c) GASNet. (d) CACDNet. (e) B2CNet. (f) ELGCNet. (g) FDFENet. (h) Farm-CD. (i) SAM-CD. (j) BAN. (k) STMTNet. (l) NeXt2Former-CD. (m) MADDRNet. (n) GT. The color legend of the detection maps is as follows: green denotes FP, red denotes FN, white denotes TP, and black denotes TN.

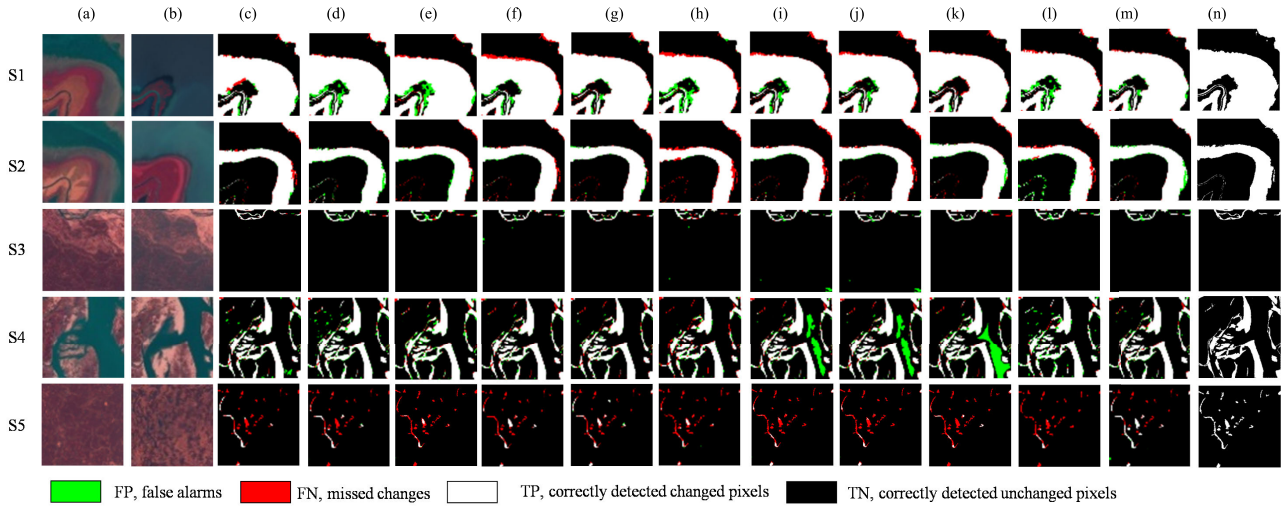


Fig. 8. Visual comparison results of the proposed method and SOTA methods on the Water-CD dataset. S1–S5 Five representative samples involving water-body changes with elongated structures, irregular boundaries, and fuzzy transition areas. (a) Image1. (b) Image2. (c) GASNet. (d) CACDNet. (e) B2CNet. (f) ELGCNet. (g) FDFENet. (h) Farm-CD. (i) SAM-CD. (j) BAN. (k) STMTNet. (l) NeXt2Former-CD. (m) MADDRNet. (n) GT. The color legend of the detection maps is as follows: green denotes FP, red denotes FN, white denotes TP, and black denotes TN.

FLOPs, MADDRNet uses 79.81M parameters and 41.13G FLOPs while achieving higher $F1$ and IoU on the Water-CD dataset. This indicates that the proposed method provides stronger detection accuracy with much lower computational complexity than large VFM-based architectures. Compared with lightweight CNN-based methods such as B2CNet and CACDNet, MADDRNet introduces a larger model scale due to the incorporation of the SkySense semantic branch. However, this additional complexity brings clear performance gains, especially in $F1$, IoU, and OA. Moreover, the SkySense branch is kept frozen during training, and only the lightweight MobileNetV2 branch, the adaptive fusion modules, and the decoder are optimized. Therefore, although MADDRNet is not the smallest model, it maintains a practical computational cost while providing more reliable detection performance in complex remote sensing scenes.

D. Training Loss Analysis

To further examine the optimization stability of MADDR-Net, Fig. 9 reports the training and validation loss curves on CLCD, GFSWCLCD, LuojiaSET-CLCD, and Water-CD. Overall, the training losses decrease rapidly in the early stage and then gradually converge to stable plateaus, indicating that the proposed semantic–morphology collaborative framework can be effectively optimized on datasets with different scene complexity. The validation losses exhibit a consistent downward trend before reaching dataset-specific plateaus, rather

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TABLE III
ABLATION STUDY OF THE DDRM, MAGF ON FOUR DATASETS

Method		Params	FLOPs	CLCD		GFSWCLCD		LuojiaSET-CLCD		Water-CD	
DDRM	MAGF	(M)	(G)	F1 (%)	IoU (%)	F1 (%)	IoU (%)	F1 (%)	IoU (%)	F1 (%)	IoU (%)
		78.73	37.40	81.63	68.96	86.07	75.55	77.72	63.56	90.52	82.69
✓		79.37	40.87	82.22	69.81	86.29	75.88	77.96	63.88	90.76	83.09
	✓	79.13	38.14	82.40	70.06	86.74	76.59	78.17	64.17	90.73	83.04
✓	✓	79.81	41.13	83.39	71.51	87.23	77.36	78.65	64.82	91.45	84.24

Note: ✓ indicates that the corresponding module is included in the model. Baseline represents the model without DDRM and MAGF modules; Red indicates the best performance.

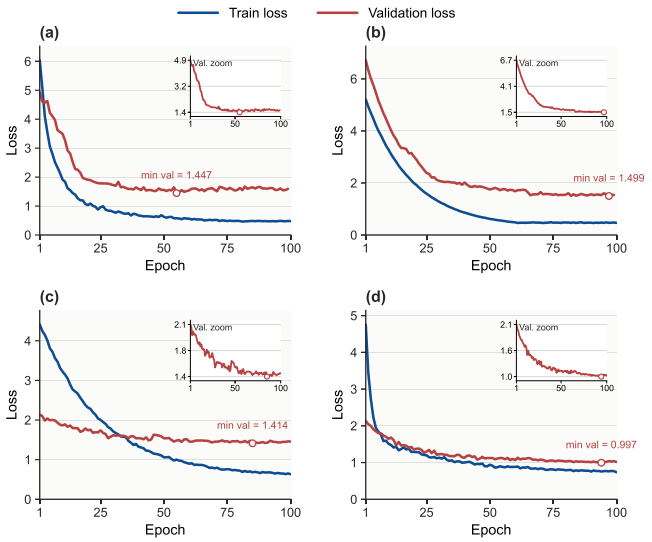


Fig. 9. Training and validation loss curves of MADDRNet on four benchmark datasets. The red hollow markers denote the minimum validation loss used for checkpoint selection. The inset in each panel provides a magnified view of the validation loss to better show its convergence behavior. (a) CLCD. (b) GFSWCLCD. (c) LuojiaSET-CLCD. (d) Water-CD.

than showing severe oscillation or divergence, which suggests stable generalization during training.

Specifically, the minimum validation losses on CLCD, GFSWCLCD, LuojiaSET-CLCD, and Water-CD are 1.447, 1.499, 1.414, and 0.997, respectively. The validation curves of CLCD and GFSWCLCD decrease sharply at the beginning and then fluctuate within a narrow range, reflecting the challenge of suppressing phenological and farmland-background interference. On LuojiaSET-CLCD, the validation loss decreases more smoothly because the dataset contains sparse small targets and heterogeneous backgrounds, where validation improvement is more gradual. For Water-CD, both training and validation losses converge to lower values, which is consistent with the strong quantitative performance reported in Table II. We select the checkpoint with the minimum validation loss for final testing, thereby avoiding the possible influence of late-stage overfitting. These observations support the convergence reliability of MADDRNet and complement the quantitative and visual comparisons reported above.

E. Ablation Study

To verify the effectiveness of MADDRNet and deeply analyze the contributions of each component (DDRM and MAGF), we conduct ablation experiments on four datasets (CLCD, GFSWCLCD, LuojiaSET-CLCD, and Water-CD), and adopt $F1$ and IoU as key evaluation metrics.

1) Module Ablation: Table III details the quantitative results after progressively integrating different modules on the four datasets. The baseline uses the "SkySense + MobileNetV2" two-stream backbone network, and incorporates from the bridge module in BAN for feature injection, without any additional enhancement modules.

a) Ablation study of DDRM: As shown in Table III, integrating DDRM increases the model parameters from 78.73M to 79.37M and FLOPs from 37.40G to 40.87G, yielding consistent performance gains across all datasets. On CLCD, $F1$ rises by 0.61%, demonstrating that DDRM effectively decouples amplitude and phase spectra, suppressing spectral style drift while preserving essential structural changes.

Adding MAGF further raises the parameters to 79.81M and FLOPs to 41.13G, producing additional $F1$ gains of 0.77%, 0.45%, 0.67%, and 0.21% on CLCD, GFSWCLCD, LuojiaSET-CLCD, and Water-CD, respectively. The module dynamically selects relevant features according to geometric morphology, suppresses nontarget responses, reduces missed detections of small targets, and improves boundary integrity, validating its effectiveness in complex scenes with fragmented farmland, irregular boundaries, and winding rivers.

b) Synergy effect: When DDRM and MAGF are jointly integrated, MADDRNet achieves consistent improvements across all four datasets, demonstrating the functional complementarity of the two modules. DDRM focuses on eliminating spectral style differences in the frequency domain to ensure semantic consistency. MAGF, on the other hand, refines the geometry in the spatial domain to ensure the integrity of the boundary. Their combination enables the model to effectively address challenges ranging from macroscopic seasonal variations to microscopic structural details.

2) Ablation Study on the Intermediate Feature Dimension: To evaluate the influence of the intermediate dimension on feature representation capability and computational burden, we conduct a parameter ablation study on the CLCD dataset by varying mid_d, as illustrated in Fig. 10. In particular, mid_d is set to 32, 64, 96, and 128 while keeping all other settings unchanged. It can be observed that increasing mid_d consistently improves the detection performance. The $F1$ rises from 80.75% to 82.41% when mid_d increases from 32 to 64, and further reaches 83.39% at mid_d = 96, indicating

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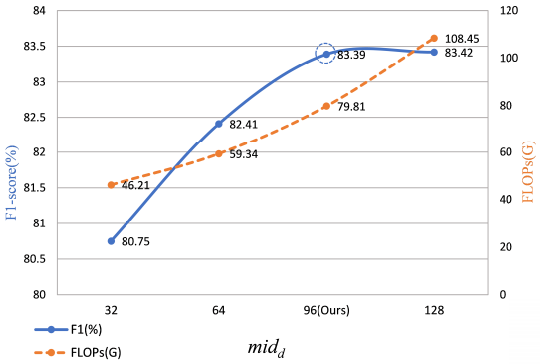


Fig. 10. Effect of the intermediate feature dimension mid_d on model performance and computational cost. Here, mid_d denotes the hidden channel dimension used in the intermediate feature transformation and fusion stage of the proposed network.

TABLE IV
GENERALIZATION PERFORMANCE OF BACKBONE ARCHITECTURES

Backbone	CLCD	
	F1	IoU
MobileNetV2	76.09	61.40
MobileNetV2+SAM	74.54	59.42
MobileNetV2+CLIP	76.47	61.90
SkySense	76.99	62.59
ResNet18+SkySense	80.35	67.16
ResNet34+SkySense	79.78	66.37
ConvNeXt+SkySense	80.74	67.70
MobileNetV2+SkySense	83.39	71.51

Red indicates the best value in each column.

that a larger intermediate dimension facilitates richer feature transformation and more discriminative change representation. However, when mid_d is further increased to 128, the $F1$ only improves marginally to 83.42%, while the FLOPs increase substantially from 79.81G to 108.45G. This suggests that excessively enlarging the intermediate dimension yields diminishing performance returns but introduces considerable computational overhead.

3) To Explore the Optimal Feature Extraction Strategy, We Compare Multiple Backbone Networks on the Clcd Dataset: The performance comparison is shown in Table IV. The combination of MobileNetV2 and SkySense significantly improved the $F1$ to 83.39% compared to SAM and CLIP [54] ($F1$: 74.54% and 76.47%). The lightweight MobileNetV2 notably outperforms the more complex ResNet and ConvNeXt [55] families (both around 80% $F1$). This shows that with the support of the SkySense VFM, the lightweight network not only avoids overfitting but also performs feature complementation more efficiently, achieving the best accuracy balance with fewer parameters.

V. DISCUSSION

In this section, we further analyze the working mechanism of MADDRNet through visualization and discuss its limitations in challenging scenarios. The visualization results reveal how the proposed framework progressively refines features from coarse semantic extraction to discriminative change enhancement and adaptive fusion.

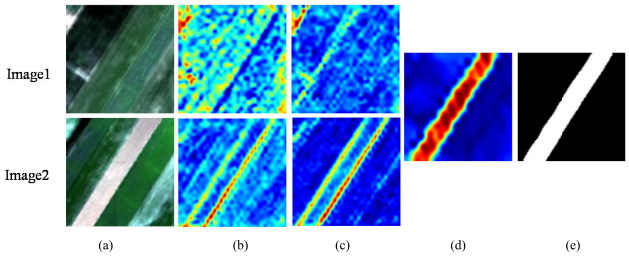


Fig. 11. Different stages are visualized in the MADDRNet. (a) Input bitemporal remote sensing images. (b) Backbone feature map. (c) DDRM feature map. (d) MAGF feature map. (e) Prediction map.

As illustrated in Fig. 11, the bitemporal input images in Fig. 11(a) show that the dominant change corresponds to a newly appeared road region in Image2. The backbone features in Fig. 11(b) capture the rough spatial morphology of this linear object, providing the semantic basis for subsequent detection. However, the activations are still relatively diffuse, with noticeable responses in the surrounding background, indicating that backbone features alone are insufficient to fully distinguish true changes from redundant interference. After DDRM processing, the responses in Fig. 11(c) become more concentrated on the actual changed region, while irrelevant background noise is effectively suppressed. This suggests that DDRM enhances discriminative change cues and reduces redundant texture responses. Based on these refined features, MAGF further performs adaptive fusion, making the representation in Fig. 11(d) sharper and more continuous along the changed road boundary. Consequently, the final prediction in Fig. 11(e) aligns well with the true shape and location of the changed region. These observations are consistent with the quantitative analysis reported in the experiments and demonstrate the complementary roles of DDRM and MAGF fusion.

Although MADDRNet achieves strong performance in handling seasonal disturbances and boundary ambiguity, certain challenges remain under extreme environmental conditions. As shown in s1 of Fig. 6, missed detections may occur when changed regions are heavily occluded by dense clouds. This is mainly because MADDRNet relies on texture and structural cues from optical imagery, which can be severely degraded under physical occlusion, thereby limiting effective feature representation.

VI. CONCLUSION

We propose MADDRNet, a novel morphology-aware dual-domain resonance framework for RSCD. In particular, DDRM mitigates spectral style drift caused by phenological variations through dual-domain collaboration, suppressing seasonal noise while preserving structural semantics. Meanwhile, MAGF combines adaptive kernel convolution with dynamic gating to alleviate boundary adhesion and fracture in complex scenes, thereby improving the integrity of change regions. Extensive experiments on four benchmark datasets,

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including CLCD ($F1$: 83.39%), GFSWCLCD ($F1$: 87.23%), LuojiaSET-CLCD ($F1$: 78.65%), and Water-CD ($F1$: 91.45%), show that MADDRNet consistently outperforms existing state-of-the-art methods in terms of $F1$ and IoU. Ablation studies further verify the effectiveness of each component. Nevertheless, missed detections may still occur under severe shadow or thick-cloud occlusion. In future work, we will explore SAR-optical fusion to improve robustness under such conditions.

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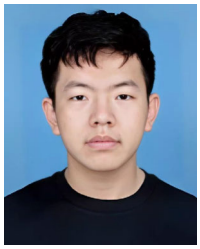
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